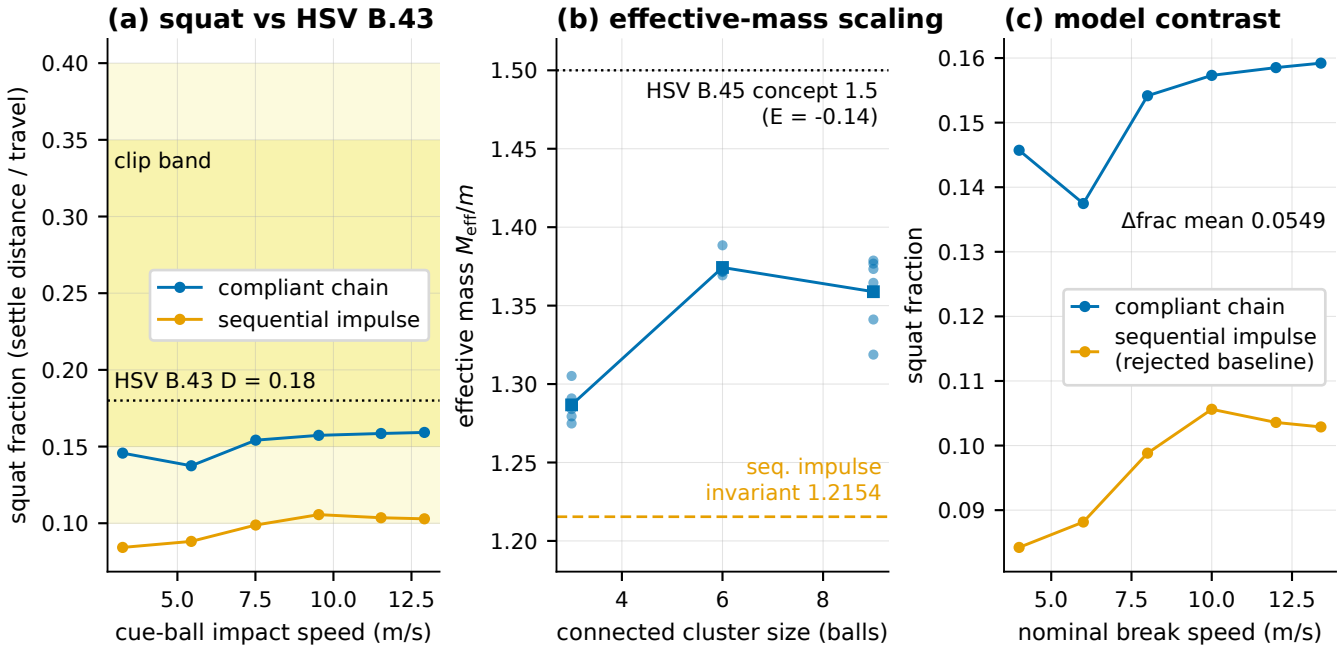
(a) squat vs HSV B.43
clip band
compliant chain
sequential impulse
HSV B.43 D = 0.18
squat fraction (settle distance / travel)
cue-ball impact speed (m/s)
(b) effective-mass scaling
HSV B.45 concept 1.5
(E = -0.14)
seq. impulse
invariant 1.2154
effective mass $M_{eff}/m$
connected cluster size (balls)
(c) model contrast
Δfrac mean 0.0549
compliant chain
sequential impulse
(rejected baseline)
squat fraction
nominal break speed (m/s)